Adaptive Frequency-Domain Active Noise Cancellation (ANC) System Using MATLAB/Simulink

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***Abstract*—Active Noise Cancellation (ANC) is a signal-processing technique used to reduce unwanted acoustic noise by generating an anti-noise signal that interferes destructively with the unwanted noise. This paper presents the design and simulation of an adaptive frequency-domain ANC system developed using MATLAB/Simulink. The proposed system accepts a desired audio signal and an unwanted noise signal as inputs. The noise component is processed in the frequency domain using the Fast Fourier Transform (FFT), followed by adaptive gain control based on error feedback. An anti-noise signal is generated and converted back into the time domain using the Inverse Fast Fourier Transform (IFFT). The anti-noise is subsequently combined with the noisy audio signal to achieve noise reduction. Simulation results are analyzed by comparing the input noisy waveform with the ANC output waveform. The proposed approach demonstrates a comparatively simple implementation of adaptive noise cancellation and provides a foundation for further development toward real-time ANC systems.**

***Index Terms*—Active Noise Cancellation, Adaptive Gain Control, FFT, IFFT, MATLAB, Simulink, Noise Reduction, Frequency-Domain Processing.**

I. Introduction

Active Noise Cancellation (ANC) is a signal-processing technique used to reduce unwanted acoustic noise by generating a secondary signal, known as anti-noise, which interferes destructively with the unwanted noise. ANC systems are widely studied for applications such as headphones, hearing aids, industrial noise control, and other acoustic environments.

Traditional ANC systems commonly use adaptive filtering algorithms such as Least Mean Squares (LMS), Normalized Least Mean Squares (NLMS), and Recursive Least Squares (RLS). These algorithms continuously adjust filter parameters according to the input and error signals to achieve effective noise reduction.

The present work focuses on the design and simulation of an adaptive frequency-domain ANC system using MATLAB/Simulink. The proposed system uses Fast Fourier Transform (FFT) and Inverse Fast Fourier Transform (IFFT) based processing along with RMS-error-based adaptive gain control. Unlike conventional LMS, NLMS, and RLS implementations, the proposed approach adjusts the gain of the generated anti-noise signal rather than updating FIR filter coefficients.

The system is developed to demonstrate the practical application of frequency-domain signal processing and adaptive control for acoustic noise reduction. The performance of the system is analyzed by comparing the noisy input signal with the resulting ANC output.

The primary objective of this work is to design and simulate an adaptive Active Noise Cancellation system using MATLAB/Simulink for reducing unwanted acoustic noise from a desired audio signal.

The objectives of the proposed system are:

- To process desired audio and unwanted noise signals.
- To analyze the noise signal using FFT-based frequency-domain processing.
- To generate an appropriate anti-noise signal.
- To implement RMS-error-based adaptive gain control.
- To reconstruct the processed signal using IFFT.
- To compare the noisy input and ANC output signals.
- To establish a simulation framework for future real-time ANC implementation.

II. Literature Review

Further studies have focused on improving the practical performance of adaptive ANC systems under different acoustic conditions. Multiple-error and multichannel adaptive filtering techniques have been investigated for applications where noise is detected at more than one spatial location. Elliott, Stothers, and Nelson introduced a multiple-error LMS approach for active control of sound and vibration, demonstrating the importance of using multiple error signals in systems with several control channels [1]. Such approaches can improve spatial control performance, although they generally require additional computational resources and more complex system architectures.

Another important aspect of practical ANC is the secondary acoustic path between the control source and the error sensor. In real systems, the generated anti-noise does not reach the error microphone instantaneously or without

modification. Therefore, secondary-path modeling is required in many conventional ANC architectures to account for the acoustic response of loudspeakers, amplifiers, and microphones. Zhang and Gan investigated online secondary-path modeling for active noise control and proposed a model-predictive approach to improve the adaptation process [2]. Similarly, Chang, Kuo, and Huang investigated secondary-path modeling for narrowband ANC systems, emphasizing its importance for reliable cancellation performance [3]. These studies indicate that accurate modeling of the acoustic path becomes increasingly important when ANC systems are implemented in practical environments.

The development of hybrid and feedback-based ANC architectures has also attracted considerable attention. Song, Gong, and Kuo presented a robust hybrid feedback ANC headset that combines feedforward and feedback control structures for headphone applications [4]. Such architectures can improve robustness when a clean reference noise signal is not always available. However, the additional feedback path and associated control requirements increase the complexity of the overall system. This creates a continuing requirement for simpler ANC structures that can demonstrate effective signal cancellation while maintaining manageable computational requirements.

Frequency-domain processing has emerged as another important direction in active noise control. Instead of performing all adaptive operations directly on individual time-domain samples, frequency-domain methods transform the signal using the DFT or FFT and perform processing on spectral components. This is particularly useful when the signal contains multiple frequency components or when relatively long filters are required. The FFT also provides a computationally efficient method for transforming a large number of time-domain samples into their frequency-domain representation [5]. However, block-based frequency-domain processing can introduce processing delay and may require additional operations for transforming signals between the time and frequency domains.

The use of subband and frequency-selective methods has also been investigated to improve ANC efficiency. Gao, Lu, and Qiu proposed a simplified subband ANC algorithm and investigated methods for reducing the computational requirements associated with secondary-path modeling [6]. Such research demonstrates that dividing the signal into frequency-related components can provide opportunities for reducing computational complexity while maintaining satisfactory cancellation performance.

Recent surveys indicate that ANC research has increasingly moved beyond basic single-channel adaptive filtering toward systems capable of handling output saturation, nonlinear behavior, multichannel operation, and changing acoustic environments. Guo *et al.* surveyed adaptive ANC algorithms addressing output saturation effects and highlighted the practical challenges associated with maintaining stable cancellation performance under actuator limitations [7]. These developments demonstrate that although conventional adaptive filters provide a strong theoretical foundation, practical ANC systems often require additional mechanisms to maintain robustness.

Adaptive-gain methods provide an alternative direction in which the amplitude of a control signal is adjusted rather than relying exclusively on continuous coefficient adaptation. Shen *et al.* investigated an adaptive-gain algorithm applied to fixed filters for active noise-control headphones [8]. The concept is particularly relevant to the present study because it demonstrates how gain adaptation can be used as part of an ANC architecture while reducing the dependence on continuously changing filter coefficients.

Virtual sensing is another research direction intended to address situations where a physical error sensor cannot conveniently be positioned at the desired control location. Moreau *et al.* reviewed virtual sensing algorithms for ANC and discussed approaches for estimating the acoustic error at locations where direct measurement is difficult [9]. Such methods are particularly relevant to practical applications where microphone placement, system size, or user comfort limits the use of conventional sensor configurations.

Wireless ANC has further expanded the scope of active noise control by using wireless reference information and look-ahead processing. Recent work presented at INTER-NOISE 2024 reviewed wireless ANC techniques and discussed frequency-domain adaptive filtering, reference-signal selection, synchronization, and computational delay [10]. These studies show that modern ANC systems must address not only the cancellation algorithm itself but also signal latency, synchronization, reference quality, and hardware constraints.

From the reviewed literature, it can be observed that conventional adaptive filtering techniques such as LMS, NLMS, and RLS remain fundamental approaches for ANC, while more recent research explores frequency-domain processing, adaptive-gain techniques, hybrid architectures, secondary-path modeling, virtual sensing, and wireless implementations. The major trade-offs involve noise-reduction capability, convergence rate, computational complexity, latency, and system robustness.

In comparison with these approaches, the present work focuses on a comparatively simple adaptive frequency-domain ANC architecture implemented in MATLAB/Simulink. The proposed system uses FFT/IFFT-based processing together with RMS-error-based adaptive gain control. The adaptation mechanism modifies the amplitude of the generated anti-noise signal according to the observed error rather than updating a conventional FIR filter coefficient vector. Therefore, the proposed work can be considered a simplified implementation that combines the frequency-domain processing concepts found in the literature with an adaptive-gain mechanism. The approach is intended primarily for simulation and conceptual validation, while advanced adaptive filtering and real-time hardware implementation remain possible directions for future development.

III. Proposed Methodology

The proposed Adaptive Active Noise Cancellation (ANC) system is developed in MATLAB/Simulink using frequency-domain signal processing and adaptive gain control. The system takes a desired audio signal and an unwanted noise signal as inputs. The noise is processed to generate an anti-noise signal, which is subsequently combined with the noisy audio signal to reduce the unwanted noise.

The overall processing sequence of the proposed system is shown in Fig. **??**.

The methodology consists of the following major stages:

A. Input Signal

A desired audio signal is provided as the primary signal. An unwanted noise signal is introduced along with the desired audio to form the noisy input signal. The signals are processed at a sampling frequency of 44.1 kHz.

B. Frequency-Domain Processing

The unwanted noise signal is transformed from the time domain into the frequency domain using the Fast Fourier Transform (FFT). This representation allows the frequency components of the noise signal to be processed independently.

C. Adaptive Gain Control

An adaptive gain-control mechanism is used to regulate the amplitude of the generated anti-noise signal. The control mechanism uses an error-related RMS measurement as feedback and adjusts the gain according to the observed error.

D. Anti-Noise Generation

The processed frequency-domain noise component is used to generate an anti-noise signal. The amplitude of this signal is controlled using the adaptive gain mechanism so that it can effectively oppose the unwanted noise component.

E. Time-Domain Reconstruction

The processed anti-noise signal is converted back to the time domain using the Inverse Fast Fourier Transform (IFFT). The real component of the reconstructed signal is then used for further processing.

F. Noise Cancellation

The generated anti-noise signal is combined with the noisy audio signal. The resulting destructive interference reduces the unwanted noise component and produces the final ANC output.

The complete processing sequence can therefore be represented as:

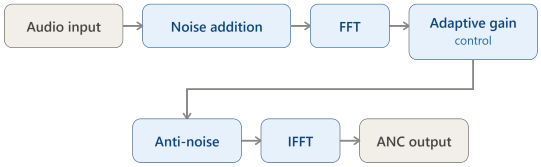


Fig. 3. Processing sequence of the proposed adaptive frequency-domain ANC system.

IV. Implementation

The proposed Adaptive Active Noise Cancellation system was implemented using MATLAB and Simulink. The complete model consists of two primary input signals: the desired audio signal and the unwanted noise signal. The audio signals are processed at a sampling frequency of 44.1 kHz.

The noise signal is initially scaled using a gain factor of 0.7 to account for the passive noise component. The desired audio signal and the scaled noise are then supplied to the Anti-Noise Generator subsystem.

Inside the Anti-Noise Generator, both the noise and the desired audio signal are converted from the time domain to the frequency domain using Fast Fourier Transform (FFT) blocks. The noise spectrum is processed using a convolution operation and an inverted frequency-domain component to generate the required anti-noise signal.

An adaptive gain-control mechanism is used to regulate the amplitude of the generated anti-noise. The RMS values of the desired signal and the feedback signal are compared to obtain an error value. This error is supplied to the MATLAB Function block, where the gain is adjusted according to the predefined error threshold.

The adaptive gain-control logic used in the implementation is given in Listing 1.

Listing 1. MATLAB function used for adaptive gain control

```
function output = fcn(err,oldgain)
req_err = 0.0001;

if(req_err<=err)
    output=oldgain+0.005;
else
    output=oldgain;
end
end
```

The generated anti-noise signal is combined with the desired audio signal and subsequently converted back to the time domain using an Inverse Fast Fourier Transform (IFFT). The real component of the reconstructed signal is extracted to obtain the final ANC output. A feedback path is also provided to continuously monitor the system response and control the adaptive gain.

The complete implementation therefore combines frequency-domain signal processing, adaptive gain control,

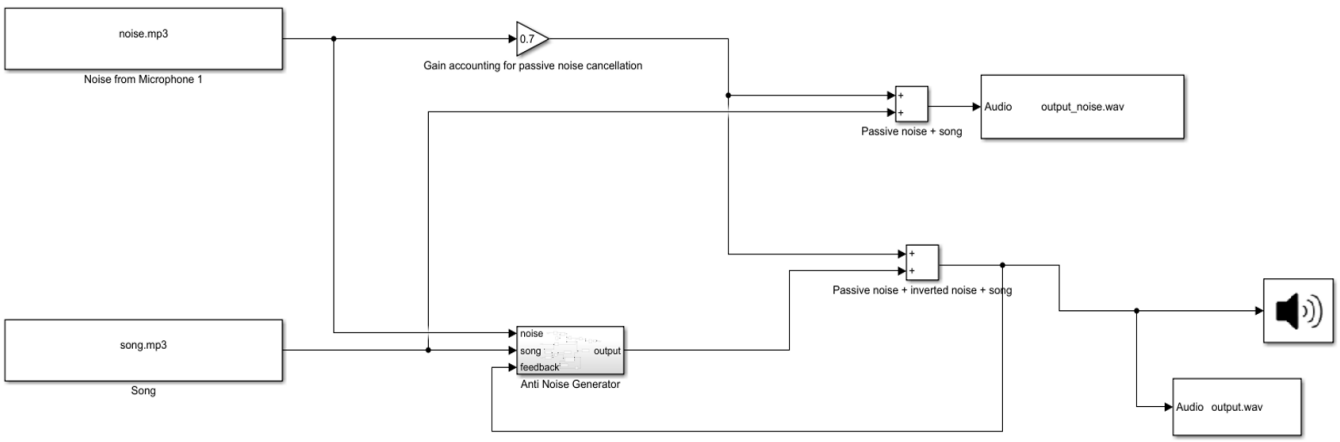


Fig. 1. Overall Simulink implementation of the proposed adaptive frequency-domain ANC system.

feedback, and time-domain reconstruction within the Simulink environment.

V. Results and Discussion

The performance of the proposed Adaptive Active Noise Cancellation system was evaluated by observing the input and output audio waveforms obtained from the MATLAB/Simulink model. The original audio signal was processed together with the unwanted noise, and the resulting ANC output was analyzed to observe the effect of the proposed noise-cancellation process.

The waveform of the original input audio signal is shown in Fig. 4.

The output waveform obtained after applying the proposed ANC process is shown in Fig. 5. The output can be compared with the original input waveform to evaluate the reduction in the unwanted noise component.

The comparison of the two waveforms provides a visual indication of the noise-cancellation performance of the proposed system. The ANC output represents the final processed audio obtained after frequency-domain processing, adaptive gain control, anti-noise generation, and time-domain reconstruction.

VI. Conclusion and Future Scope

The proposed adaptive frequency-domain Active Noise Cancellation (ANC) system was successfully designed and simulated using MATLAB/Simulink. The system combines frequency-domain signal processing using FFT/IFFT with RMS-error-based adaptive gain control to generate an anti-noise signal. The processed output was analyzed by comparing the input audio signal with the ANC output waveform.

The developed model demonstrates the practical application of adaptive signal processing for acoustic noise reduction and provides a relatively simple framework for understanding the operation of an ANC system. The MATLAB/Simulink implementation also provides flexibility for further modification and performance evaluation.

Future work can focus on improving the adaptive algorithm and extending the system toward real-time operation. Conventional adaptive filtering techniques such as LMS, NLMS, RLS, and Filtered-X LMS (FxLMS) can be investigated for improved convergence and noise-cancellation performance. The system can also be extended to microphone-based real-time audio input, speaker-based anti-noise output, multi-channel ANC, secondary-path modeling, and hardware implementation using embedded DSP or other real-time processing platforms.

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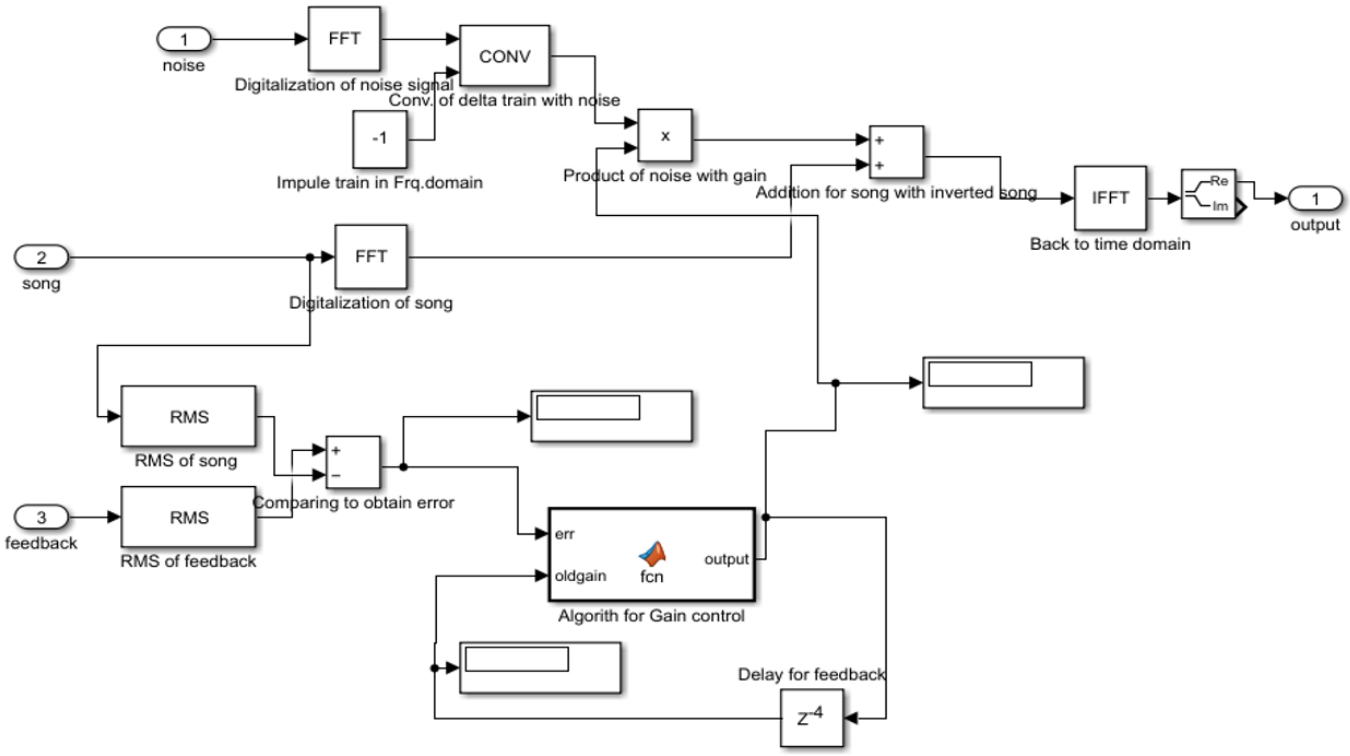


Fig. 2. Internal structure of the Anti-Noise Generator and adaptive gain-control algorithm.

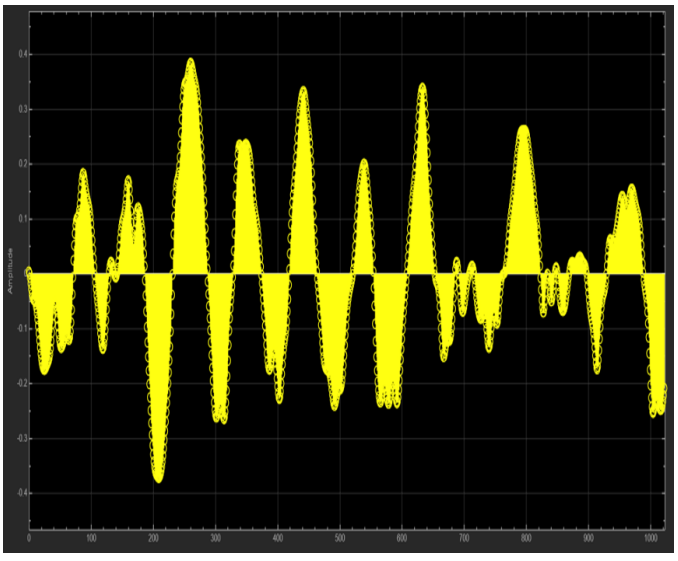


Fig. 4. Waveform of the original input audio signal.

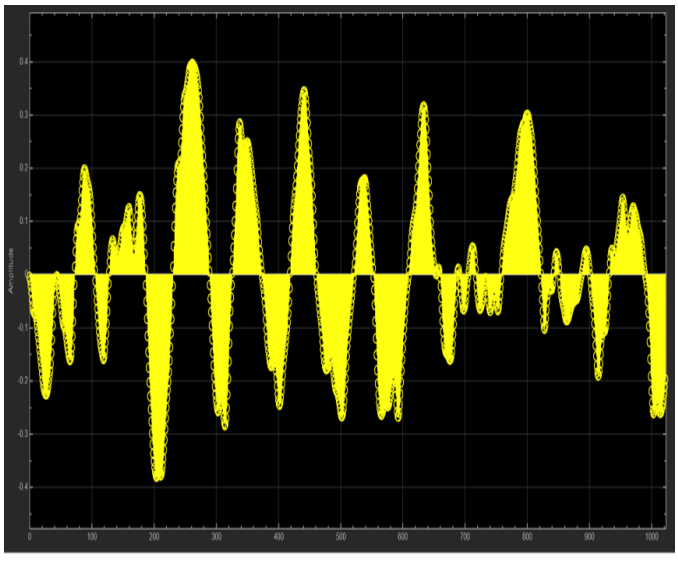


Fig. 5. Waveform of the audio signal after ANC processing.

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